

CLASS 1 · CAPSULE 1 · TECHNICAL APPENDIX

Why AI succeeds or fails on each Frontier Check task

The mechanisms, numbers, and sources behind the eight-task Frontier Check — a technical appendix for students who want the exposition.

This appendix is optional and not examinable. It is for the students — often, though not only, from a computing or engineering background — who look at the eight tasks and want to know precisely why a system that wins a mathematics olympiad cannot reliably read a clock. Each task pairs the headline result with the underlying mechanism: how the model represents the input, what training signal was available, what the task actually requires as a computation, and where that computation has no counterpart in the architecture. The recurring lesson is that the difficulty a task presents to a human is nearly uncorrelated with its difficulty for a current model — Mollick's *jagged frontier*.

THE PATTERN UNDERNEATH THE EIGHT

One architectural fact organises most of what follows. A modern large language model is a decoder-only transformer: a function that maps a sequence of tokens to a probability distribution over the next token, $p(x_t | x_1 \dots x_{t-1})$, trained by maximum likelihood (cross-entropy) on that next-token objective. It has no persistent external memory and no named variables — its only working state is the activation pattern recomputed over the context window at each step. Any task that needs exact, updatable discrete state — a counter, a position index, a running length — must therefore simulate that state inside the token stream itself, because the network provides no register for it. Capability concentrates where three properties hold; it collapses where their opposites do.

Where AI succeeds	Where it fails
The objective is verifiable — an answer can be checked mechanically. This is what unlocks reinforcement learning against a reliable reward, and search at inference time (generate many candidates, verify, keep what survives).	The task needs exact discrete state held across a process — counting, position, length. There is no register for this in a next-token predictor; it must be emulated in-context or offloaded to a tool.
Data is abundant and well-structured , often because some institution spent decades curating it (the PDB, competition-maths corpora, case law).	The state is continuous, deformable and partially observed — the physical world, where data cannot be scraped at internet scale.
The output space is bounded and well-defined , so plausibility and correctness nearly coincide.	The long tail dominates — rare cases are, by definition, thinly represented, and are often exactly the ones that matter.

What is absent from both columns is how hard the task feels to a human. That omission is the whole claim of the capsule.

THE EIGHT, ONE BY ONE

ACES 1 · Win a gold medal at the International Mathematical Olympiad

Result. Gemini Deep Think scored 35 of 42 points — at or above the gold-medal threshold — at the 2025 IMO, reasoning end to end in natural language within the 4.5-hour limit, up from a 28-point silver in 2024. An OpenAI reasoning model reported a comparable result the same year.

- **Verifiable objective** → **RLVR**. A proof or final answer is checkable, so the reward is a near-exact 0/1 signal rather than a learned approximation. This is reinforcement learning from verifiable rewards (RLVR): it sidesteps the reward-model error and reward-hacking that limit ordinary RLHF, and gives a clean gradient toward correct reasoning.
- **Inference-time compute (System 2)**. This is not one forward pass. The model spends test-time compute — long chains of reasoning, parallel sampling of many solution paths, self-verification and revision — trading latency for accuracy. Karpathy framed this as the missing 'System 2'; scaling it is a distinct axis from scaling parameters or data.
- **Dense, well-structured signal**. Competition mathematics is unusually well documented — decades of problems, worked solutions, and formal proof corpora — so the pre-training distribution already covers the genre densely.

The honest caveat: IMO problems are hard but closed — a definite answer, a fixed format, no ambiguity about what counts as success. Very little real-world work has that shape, which is why olympiad performance transfers less than the headline implies.

FAILS 2 · Read an analog clock — 50.6% vs 90.1% for humans

Benchmark. ClockBench, reported in the AI Index 2026: the best model reads an analog clock correctly 50.6% of the time; humans manage 90.1%.

- **Representation mismatch.** A vision transformer splits the image into fixed patches (e.g. 14×14 or 16×16 pixels), linearly embeds each, and adds positional embeddings. Sub-patch geometry — the precise angle of a thin hand — is averaged inside a patch and never preserved as an explicit scalar the later layers can read.
- **No geometric primitive.** The task is really a small pipeline: locate two hands, estimate their angles θ_{hour} and θ_{minute} , then map those to a time by modular arithmetic. The network has no atan2, no angle-measuring operation; it must recover the answer associatively from visual features — a fundamentally different computation from the one the task needs.
- **Thin, skewed training data.** Analog clocks are rare in image–caption pairs, and captions almost never state the exact time. There is also a well-known convention of photographing watches at roughly 10:10 to frame the brand — which would bias any model trained on scraped commercial imagery. Plausible and frequently cited; treat as hypothesis, not established finding.

This is the flagship example: the same class of system that produced olympiad-standard mathematics cannot reliably do what a six-year-old does. Difficulty and capability are unrelated axes.

ACES 3 · Predict a protein's 3D structure

Result. AlphaFold 2 effectively solved a fifty-year problem at CASP14 (median GDT $\approx$ 92). AlphaFold 3 (Abramson et al., Nature 630, 493–500) generalises to complexes — proteins with nucleic acids, ligands, ions and modified residues — by replacing the AF2 structure module with a diffusion module that denoises directly on atomic coordinates.

- **Enormous curated labelled data.** The Protein Data Bank is the accumulated output of decades of experimental crystallography and cryo-EM — roughly 200k structures of institutionally curated ground truth, not scraped text.
- **A strong physical prior.** Multiple-sequence alignment supplies co-evolutionary signal: residues that mutate together across evolution tend to be in spatial contact. Evolution effectively ran the experiment and left the answer in the sequence record; the Evoformer turns that into a pairwise contact representation.
- **Bounded, checkable output.** The answer is a set of 3D coordinates scored against known structures (GDT, IDDT, TM-score) — verifiable, exactly as in task 1.

The recurring trio: curated data + a verifiable objective + a bounded output space — and nothing to do with the task being 'easy'. The honest limits remain: disordered regions, large conformational change, and ligand poses without homologous precedent are still weak spots.

FAILS 4 · Count the letters in a word

Mechanism — the one that matters most for this course. Text is segmented by byte-pair encoding (BPE) into subword tokens and converted to integer IDs before the model sees anything. 'strawberry' becomes roughly three tokens — something like str / aw / berry $\rightarrow$ [496, 1403, 19772] — and each ID maps to an embedding vector. The characters inside a token are not exposed to the network at all.

- **No character-level index.** There is no operation that enumerates characters within a token. Whatever spelling knowledge the model has was learned incidentally — from hyphenation, wordplay, acrostics in the training text — so it is patchy and unreliable. The same root causes digit-level arithmetic errors and difficulty reversing strings.
- **Why '2' specifically.** The double-r in -berry is a salient, frequently discussed spelling feature, and berry / cherry / merry all have exactly two r's. The model retrieves that association; the lone r inside the str token is the boundary case that gets dropped.
- **Two fixes, both instructive.** Give it a tool (call a function that counts characters), or make it spell the word out first in a reasoning trace — which converts 'recall a property of an opaque token' into 'read a list I just wrote'. This is why reasoning models frequently now pass, and why a live demo may fail to fail.

If you run it yourself: 'strawberry' is now famous enough to be over-represented in training data. Words like chrysanthemum or onomatopoeia break the behaviour more reliably.

ACES — with an asterisk 5 · Pass a professional law exam

Result, and the contested part. GPT-4 passed the Uniform Bar Exam; OpenAI reported roughly 90th percentile at launch. Martínez (2024, Artificial Intelligence and Law) re-analysed the claim and found the comparison group was heavily skewed toward repeat takers who had already failed. Against first-time takers the estimate falls to ≈62nd percentile overall and ≈42nd on the written essays; against those who actually passed, ≈48th overall and ≈15th on essays.

- **Why it passes at all.** The multiple-choice component rewards retrieval and pattern application over a large, well-documented, heavily represented corpus — statutes, case law, and an entire commercial test-prep industry. Ranking answer plausibility is enough, and plausibility is what the model is built to estimate.
- **Why the essays are weakest.** Essays require sustained, structured argument graded on reasoning rather than recall — precisely the dimension where plausibility substitutes poorly for correctness, and where the percentile collapses.
- **The transferable lesson.** How a benchmark is scored, and against which comparison group, can move a headline number by 40 percentile points. Read every 'AI passed X' claim by asking what the denominator was.

FAILS 6 · Fold a pile of laundry — 12% vs 89.4% in simulation

Benchmark. Robots succeed at roughly 12% of real household tasks; on RLBench, in software simulation, manipulation reaches 89.4%. That gap is the current state of physical AI.

- **Deformable, partially observed objects.** Cloth has effectively unbounded configuration degrees of freedom, self-occludes constantly, and is only partially observable. A rigid cube has a pose; a crumpled shirt does not — there is no low-dimensional state to estimate.
- **The sim-to-real gap.** No simulator models cloth well enough to transfer: friction, drape, and fold memory (hysteresis) are exactly what simulation approximates worst — and simulation is where the 89.4% comes from. Domain randomisation narrows but does not close the gap.
- **Data cannot be scraped.** There is no internet-scale corpus of robot manipulation. Every demonstration is physically collected in real time; even pooled efforts like Open X-Embodiment are minuscule beside text corpora.
- **Failure is unrecoverable.** A wrong token can be regenerated; a dropped plate cannot. That asymmetry forces conservatism, redundancy and safety margins into every real-world system, which further lowers measured success rates.

FAILS 7 · Write a poem of exactly 50 words

Mollick's own example in Co-Intelligence: the model produces a genuinely good poem — of the wrong length.

- **Same tokenisation root as task 4.** It is not counting words, because words are not the unit it operates on; tokens are.
- **No lookahead, no length register.** Autoregressive generation commits to each token in turn with no plan for the whole. Hitting an exact length needs a running count steered toward a target — state the architecture does not maintain. External constrained decoding or a tool can impose it; unaided generation cannot.
- **The instructive contrast.** The model writes an excellent sonnet, which has far more constraints — fourteen lines, a rhyme scheme, iambic pentameter. Form is stylistic pattern, learnable from examples; exact length is arithmetic. Different problem classes, and only one is hard for this architecture.

IT DEPENDS 8 · Drive a car through a city

Both answers are true. Waymo runs roughly 450,000 paid trips a week across several US cities; in China, Apollo Go has completed more than 11 million fully driverless rides, up about 175% year on year. And it remains unsolved in the general case.

- **Operational design domain (ODD).** The honest unit of analysis is not 'can it drive' but 'within what envelope' — which cities, mapped to what resolution, which weather and speeds, with what remote human assistance on call. Every deployed system has a defined ODD, and its performance is only claimed inside it.
- **The long tail bounds safety.** Rare events are under-represented in training data and are precisely the safety-critical ones. Aggregate performance is set by the tail, not the average — which is why 'mostly works' is not the relevant bar.

- **Why the card has no correct box.** Deliberate. The question it forces — under what conditions? — is the most useful one in the course, and C5 returns to it as the physical-AI case.

ON THE TAXONOMY (A TECHNICAL ASIDE)

A careful reader will notice the six 'types' of AI used in class are not a clean architectural taxonomy — and that is correct. Foundation model is a scale-and-training category while generative is an output category, so they overlap heavily. An agent is not a model type at all but an architectural pattern — a foundation model plus tools plus a control loop. Multimodal is a property of inputs and outputs. Physical / embodied AI is a deployment domain. Of the set, only discriminative ML is a genuinely distinct model family.

The categories are deliberately a practitioner's taxonomy, organised by what job a system does and how it fails, not a computer scientist's taxonomy organised by architecture. For choosing a tool for a business problem, 'how does this fail and who does it hurt' is a more useful axis than 'what is the model family' — which is why the course keeps it.

SOURCES

- Stanford HAI, AI Index Report 2026 — ClockBench, RLBench, OSWorld, Waymo and Apollo Go figures; Takeaway #4 ('jagged intelligence').
- Mollick, E. (2024). Co-Intelligence. Portfolio — the jagged frontier (Ch. 3, pp. 47–48); the fifty-word poem.
- Abramson, J. et al. (2024). 'Accurate structure prediction of biomolecular interactions with AlphaFold 3.' Nature 630, 493–500.
- Jumper, J. et al. (2021). 'Highly accurate protein structure prediction with AlphaFold.' Nature 596 — AF2 / CASP14.
- Martínez, E. (2024). 'Re-evaluating GPT-4's bar exam performance.' Artificial Intelligence and Law.
- Karpathy, A. (2023). 'Intro to Large Language Models' — tokenisation, the two-file model, System 1 vs System 2.

PROVENANCE & CAVEATS

Generated and fact-checked using Anthropic models against the AI Index 2026 PDF, the AlphaFold 3 paper on Nature.com, and Martínez (2024). Two claims are explicitly flagged as hypothesis rather than established finding: the analog-clock 10:10 photography skew, and the specific token boundaries shown for 'strawberry' (which vary by tokeniser). Benchmark numbers are point-in-time and will move. Errors remain possible — flag anything that looks wrong.